

SLIP – 5

Q 1) Read the data from file 'student.txt' containing names of student and their class(FY,SY,TY). Accept a student name from user and use binary search algorithm to check whether the name is present in the file and output the class of student otherwise output “Student not in the list”.

Assumed file format (one record per line): Name Class e.g. Amit FY Assumption: student.txt is sorted alphabetically by student name.

Answer:

```
#include <stdio.h>
#include <string.h>

/* Assumption: student.txt is sorted alphabetically by student name */ struct
Student
{
    char name[50];
    char cls[5];
};

int main()
{
    FILE *fp;    struct Student s[100];    char
    key[50];    int n = 0, low, high, mid, cmp,
    found = 0;

    fp = fopen("student.txt", "r");
    if(fp == NULL)
    {
        printf("File not found");
        return 0;
    }

    while(n < 100 && fscanf(fp, "%s %s", s[n].name, s[n].cls) == 2)
        n++;    fclose(fp);

    printf("Enter student name: ");
    scanf("%s", key);

    low = 0;
    high = n - 1;

    while(low <= high)
    {
```

```

        mid = (low + high) / 2;
    cmp = strcmp(s[mid].name, key);

    if(cmp == 0)
    {
        printf("Class of %s is %s", key, s[mid].cls);
        found = 1;
        break;
    }
    else if(cmp
    < 0)
        low = mid + 1;
    else
        high = mid - 1;
    }

    if(!found)
        printf("Student not
    in the list");

    return 0;
}

```

Section II: Database Management Systems [15 Marks]

Q.2) Consider the following database (primary keys are underlined).

Doctor (dno, dname, city)

Patient (pno, pat_name, city, disease)

The relationship is as follows: Doctor-Patient: many-to-many with Fees as a descriptive attribute.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```

CREATE TABLE Doctor(
    dno INT PRIMARY KEY,
    dname VARCHAR(50),
    city VARCHAR(50)
);

```

```

CREATE TABLE Patient(
    pno INT PRIMARY KEY,
    pat_name VARCHAR(50),
    city VARCHAR(50),
    disease VARCHAR(50)
);

```

```
CREATE TABLE Doctor_Patient(      dno
INT REFERENCES Doctor(dno),      pno
INT REFERENCES Patient(pno),      fees
NUMERIC(8,2),
PRIMARY KEY(dno, pno)
);
```

```
INSERT INTO Doctor VALUES
(1,'Dr. Patil','Nagpur'), (2,'Dr.
Kumar','Pune');
```

```
INSERT INTO Patient VALUES
(1,'Asha','Pune','Asthama'),
(2,'Anil','Nagpur','cancer'),
(3,'Bala','Pune','Asthama');
```

```
INSERT INTO Doctor_Patient VALUES
(1,2,500),
(2,1,200),
(2,3,200);
```

Q 2(B)

B. Execute the following queries in PostGreSQL(any 3) [6 marks]

i. List the details of all the doctors from 'Nagpur' city.

Answer:

```
SELECT * FROM Doctor
WHERE city = 'Nagpur'; ii.
```

Display count of patients.

Answer:

```
SELECT COUNT(*) FROM Patient;
```

iii. List the names of the patients suffering from 'cancer' disease.

Answer:

```
SELECT pat_name FROM Patient WHERE
disease = 'cancer';
```

iv. Change the city of 'Dr. Patil' to Pune.

Q 2(C)

C. Write a query to find the number of patients suffering from “Asthama” and have been charged a fee of Rs. 200 by “Dr. Kumar”. [4 marks]

Answer:

```
SELECT COUNT(*)
FROM Doctor_Patient DP
JOIN Doctor D ON DP.dno = D.dno
JOIN Patient P ON DP.pno = P.pno
WHERE P.disease = 'Asthama'
AND DP.fees = 200
AND D.dname = 'Dr. Kumar';
```

Q 3) Viva [5 marks]